

HYEONG JIN HYUN

Department of Statistics and Data Science, University of Cincinnati

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EMPLOYMENT

University of Cincinnati	2026 – Present
• Assistant Professor • Department of Statistics and Data Science	
Purdue University	2025 – 2026
• Postdoctoral Scholar • Principal Investigators: Dr. Xiao Wang, Dr. Haibo Liu	

EDUCATION

Purdue University	U.S.A, 2025
• <i>Ph.D. in Statistics</i>, Advisor: Dr. Xiao Wang	
Seoul National University	South Korea, 2020
• <i>M.S. in Statistics</i>, Advisor: Dr. Johan Lim	
Seoul National University	South Korea, 2018
• <i>B.S. in Mathematics and Statistics (double major)</i>, <i>Cum Laude</i> • Leave of absence for military service (2014 - 2016)	

RESEARCH INTERESTS

Simulation-based inference (SBI), approximate Bayesian computation (ABC), conformal prediction, generative model, machine learning theory, uncertainty quantification, physics informed neural network (PINN)

PUBLICATIONS & SUBMITTED MANUSCRIPTS

(* indicates equal first authorship)

- **Hyun, H. J.**, & Wang, X. (2026+). *Neural Amortized Bayesian Computation*. *Submitted*.
- Liu, Y., **Hyun, H. J.**, & Wang, X. (2026). *Regularized Physics-Informed Neural Networks for Parameter Estimation in Differential Equation Models*. **Technometrics**.
- **Hyun, H. J.**, & Wang, X. (2025). *Neural Conformal Inference for Jump Diffusion Processes*. **Journal of Econometrics**, 251, 106061.
Award: 2023 ASA Best Student Paper Award (Business & Economics Section).
- **Hyun, H. J.**, & Wang, X. (2024). *Fast Cost-Constrained High-Dimensional Regression*. **Statistica Sinica**. *In press*. https://www3.stat.sinica.edu.tw/ss_newpaper/SS-2024-0331_na.pdf.
Award: 2024 ICOSA Best Poster Award.
- **Hyun, H. J.***, Kim, Y.*, Kim, S. J., Kim, J., Lim, J., Lim, D. K., & Kwon, S. W. (2021). *Constrained Principal Component Analysis with Stochastically Ordered Scores for High-Dimensional Mass Spectrometry Data*. **Chemometrics and Intelligent Laboratory Systems**, 216, 104376.

- Kim, J.^{*}, **Hyun, H. J.^{*}**, Choi, E. A., Kim, Y., Bae, Y. J., & Kang, H. T. (2020). *Metformin Use Reduced the Risk of Stomach Cancer in Diabetic Patients in Korea*. **Gastric Cancer**, 23(6), 1075–1083.
- Kim, J., Choi, E. A., Han, Y. E., Lee, J. W., Kim, Y. S., Kim, Y., You, H. S., **Hyun, H. J.**, & Kang, H. T. (2020). *Association Between Statin Use and All-Cause Mortality in Cancer Survivors*. **Nutrition, Metabolism and Cardiovascular Diseases**, 30(3), 434–440.

WORKING PAPERS

- (**Hyun, H. J.**, Wang, X.) Sequential Monte Carlo amortized neural Bayesian computation.
- (Shin, H., **Hyun, H. J.**, Wang, X.) Simulation-based inference approaches for Hawkes processes with earthquake data.
- (**Hyun, H. J.**, Liu, H., Wang, X.) Neural posterior estimation with amortized calibration.

TEACHING EXPERIENCE

- Instructor
 - STAT 598: Qualifying exam preparation course Purdue, 2024-2026
Designed and taught a preparatory course in mathematical statistics for first-year Ph.D. students.
Provided detailed solutions to past 15 years of qualifying exams to enhance student readiness.
- Teaching Assistant
 - STAT 301: Elementary Statistical Methodology Purdue, 2020-2023
Led weekly lab sessions to teach statistical methods and demonstrated SPSS functionality.
Designed, distributed, and graded lab assignments to enhance student learning.
 - STAT 519: Introduction to Probability Theory Purdue, 2022
Provided personalized guidance during weekly office hours.
 - STAT 512: Applied Regression Analysis Purdue, 2020
Held lab sessions to learn how to implement regression techniques using R.
 - Mathematical Statistics SNU, 2018
Held weekly six-hour recitation, teaching problem-solving strategies for exercises.

AWARDS

2025 William J. Studden Publication Award	Purdue, April 2025
2024 ICSA Best Poster Award	ICSA, June 2024
2023 ASA Best Student Paper Award B&E Section	ASA, August 2023
2023 ICSA Best Student Paper Award Honorable Mention	ICSA, June 2023

TALKS

2026 JSM Topic-Contributed Session	Boston, August, 2026
Invited Talk	University of Cincinnati, January, 2026
2024 Purdue GSO	West Lafayette, November 2024
2024 JSM Topic-Contributed Session	Portland, August 2024
2024 ICSA Poster Session	Nashville, August 2024
Purdue GSO Seminar	West Lafayette, November 2023
2023 ICSA Symposium Best Student Paper Award Presentation	Ann Arbor, June 2023
2023 JSM Best Student Paper Award Presentation	Toronto, August 2023

SKILLS & LANGUAGES

Languages	Korean (native), Jejueo (native), English (fluent)
Computation Skills	Python, R, MATLAB, SPSS

SOFTWARE

- NCoin-JDP: <https://github.com/anonymous1116/NCoin-JDP>
- FCR: <https://github.com/anonymous1116/FCR>
- os-PCA: <https://github.com/hjhhyu0081/osPCA>